INCH-POUND
MIL-DTL-32625
23 January 2019
SUPERSEDING
MC/PD 06 2011 SYSCOM
30 October 2011

DETAIL SPECIFICATION

PULLOVER SHIRT, FLAME RESISTANT, MIDWEIGHT, COLD WEATHER

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a flame resistant (FR), midweight, cold weather pullover shirt. The pullover shirt will be one type and in the following sizes and lengths, as specified (see 6.2).

Extra Small	Small	Medium	Large	Extra Large
Regular	Regular	Regular Long	Regular Long	Regular Long

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

COMMERCIAL ITEM DESCRIPTIONS

A-A-55195 - Thread, Para-Aramid, Spun, Intermediate Modulus

A-A-55217 - Thread, Aramid, Spun Staple

A-A-55634 - Zippers (Fasteners, Slide Interlocking)

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-32075 - Label: For Clothing, Equipage, and Tentage, (General Use)

(Copies of these documents are available online at https://quicksearch.dla.mil/.)

Comments, suggestions, or questions on this document should be addressed to Marine Corps Systems Command, 2200 Lester Street, Quantico, VA 22134 ATTN: SEAL-SE-STDS or emailed to USMC_STDZ@usmc.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 8415

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2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN ASSOCIATION OF TEXTILE CHEMISTS AND COLORISTS (AATCC)

AATCC EP9	-	Visual Assessment of Color Difference of Textiles
AATCC TM8	-	Colorfastness to Crocking: Crockmeter Method
AATCC TM16.3	-	Colorfastness to Light: Xenon-Arc
AATCC TM20A	-	Fiber Analysis: Quantitative
AATCC TM61	-	Colorfastness to Laundering: Accelerated
AATCC TM135	-	Dimensional Changes of Fabrics after Home Laundering

(Copies of these documents are available online at www.aatcc.org.)

AMERICAN SOCIETY FOR QUALITY (ASQ)

ASQ/ANSI Z1.4	-	Sampling Procedures and Tables for Inspection by Attributes

(Copies of this document are available online at www.asq.org.)

ASTM INTERNATIONAL

ASTM D737	-	Standard Test Method for Air Permeability of Textile Fabrics
ASTM D1577	-	Standard Test Methods for Linear Density of Textile Fibers
ASTM D1776/D1776M	-	Standard Practice for Conditioning and Testing Textiles
ASTM D1777	-	Standard Test Method for Thickness of Textile Materials
ASTM D2594	-	Standard Test Method for Stretch Properties of Knitted Fabrics Having Low Power
ASTM D3512/D3512M	-	Standard Test Method for Pilling Resistance and Other Related Surface Changes of Textile Fabrics: Random Tumble Pilling Tester
ASTM D3776/D3776M	-	Standard Test Methods for Mass Per Unit Area (Weight) of Fabric
ASTM D3787	-	Standard Test Method for Bursting Strength of Textiles–Constant-Rate-of-Traverse (CRT) Ball Burst Test
ASTM D5034	-	Standard Test Method for Breaking Strength and Elongation of Textile Fabrics (Grab Test)
ASTM D5103	-	Standard Test Method for Length and Length Distribution of Manufactured Staple Fibers (Single-Fiber Test)
ASTM D6193	-	Standard Practice for Stitches and Seams
ASTM D6413/D6413M	-	Standard Test Method for Flame Resistance of Textiles (Vertical Test)

(Copies of these documents are available online at www.astm.org.)

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CRC PRESS

Hayes, A. Wallace (editor) and Kruger, Claire L. (editor). *Hayes' Principles and Methods of Toxicology*. 1989. 394-396.

(There are a number of ways through which this book may be obtained, such as online retailers, bookstores, public library, to name a few.)

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

ISO 11092 - Textiles – Physiological Effects – Measurement of Thermal and Water-Vapour Resistance Under Steady-State Conditions (Sweating Guarded-Hotplate Test)

(Copies of this document are available online at www.iso.org.)

JOHN WILEY & SONS INC

Marzulli, Francis and Maibach, Howard. "Contact Allergy: Predictive Testing in Humans." *Advances in Modern Toxicology: Dermatoxicology and Pharmacology, Volume 4*. John Wiley & Sons Inc. 1977. 353-372.

(Copies of this document are available from the U.S. Army Center for Health Promotion and Preventative Medicine, ATTN: MCHB-DC-TTE, Bldg., E-2100, Aberdeen Proving Ground, MD 21010-5422.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Inspections.

3.1.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.1.2 Conformance inspection. When specified (see 6.2), a sample shall be subjected to a conformance inspection in accordance with 4.3.

3.2 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.3 Materials and components.

3.3.1 Materials.

3.3.1.1 Basic material. The basic material for the pullover shirt shall be a FR plaited knit fabric, constructed with a modacrylic/polyester/rayon/spandex blend, Polartec style 91371, or equal. The color of the fabric shall be Coyote 498 matching the reference standard furnished by the Government (see 6.2). The fabric shall show no toxicity when used as intended (see 6.1) and when tested as specified in 4.4.5. The fabric shall conform to the physical requirements specified in table I when tested as specified in 4.4.1.

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TABLE I. Basic material physical requirements.

Material Characteristic	Material Requirement
Construction	Plaited knit with face to back yarn denier gradient 1:2 differential
Fiber content	Modacrylic/polyester/rayon/spandex blend
Weight	7.625±0.625 oz./yd^2
Air permeability (minimum)	120 cubic feet/square feet/minute
Bursting strength (minimum)	40 pounds
Colorfastness (minimum)	
Laundering (3 cycles)	4.0
Crocking	
Wet	3.0
Dry	4.0
Light, Xenon	4.0
Dimensional stability (3 cycles) (maximum)	
Warp	5.0%
Filling	5.0%
Fabric stretch (minimum)	
Wales	35%
Courses	60%
Pilling (jersey face) (minimum)	
Initial	4.0
After laundering (3 cycles)	4.0
Thermal insulation (Clo) (minimum)	
Initial	0.3
After laundering (3 cycles)	0.25
Thickness	0.06 to 0.08 inch at 0.06 pounds per square inch
Vertical flame (initial) Wales and courses (maximum)	
After flame	2.0 seconds
After glow	0 seconds
Char length	5.0 inches
Melt/drip	No melt/no drip
Vertical flame (after 5 launderings) Wales and courses (maximum)	
After flame	2.0 seconds
After glow	0 seconds
Char length	5.0 inches
Melt/drip	No melt/no drip
Visual shade matching	
Daylight D75 (7500±200K)	Good match to standard sample
Incandescent (2856±200K)	Good match to standard sample

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3.3.1.2 Under collar, zipper protective flap backing, and lower front facings. The material for the under collar, zipper protective flap backing, and lower front facings shall be a plain weave aramid cloth; the fiber content shall be 95 percent meta-aramid and 5 percent para-aramid. The color shall be Coyote 498, matching the reference standard furnished by the Government (see 6.2). The fabric shall show no toxicity when used as intended (see 6.1) and when tested as specified in 4.4.5. The material shall conform to the requirements in table II when tested as specified in 4.4.1.

TABLE II. Under collar, zipper protective flap backing, and lower front facing material requirements.

Material Characteristic	Material Requirement
Construction	Plain weave
Fiber content	95% Meta-aramid/5% Para-aramid
Weight	4.5±0.5 oz./yd^2
Fiber denier	1.5 denier per filament
Fiber staple length	1½ to 2 inches
Yarn type	
Warp	2 ply spun
Filling	2 ply spun
Breaking strength	
Warp	125 pounds
Filling	90 pounds
Char temperature (minimum)	675 °F
Colorfastness (minimum)	
Laundering (3 cycles)	4.0
Crocking	
Wet	3.0
Dry	4.0
Light, Xenon	3.0
Dimensional stability (3 cycles) (maximum)	
Wales	3.5%
Courses	3.5%
Vertical flame (initial) Warp and fill (maximum)	
After flame	2.0 seconds
After glow	10 seconds
Char length	5.0 inches
Melt/drip	No melt/no drip
Vertical flame (after 5 launderings) Warp and fill (maximum)	
After flame	2.0 seconds
After glow	10 seconds
Char length	5.0 inches
Melt/drip	No melt/no drip
Visual shade matching	
Daylight D75 (7500±200K)	Good match to standard sample
Incandescent (2856±200K)	Good match to standard sample

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3.3.1.3 Pocket lining. The pocket lining for the pullover shall be an FR jersey knit fabric constructed with a modacrylic/polyester/rayon/spandex blend, Polartec style 2010, or equal. The color of the fabric shall be Coyote 498 matching the reference standard furnished by the Government (see 6.2). The fabric shall show no toxicity when used as intended (see 6.1) and when tested as specified in 4.4.5. The fabric shall conform to the physical requirements specified in table III when tested as specified in paragraph 4.4.1.

TABLE III. Pocket lining requirements.

Material characteristic	Material requirement
Construction	Jersey knit
Fiber content	Modacrylic/polyester/rayon/spandex blend
Weight	6.75 to 8.25 oz./yd^2
Bursting strength (minimum)	40 pounds
Colorfastness (minimum)	
Laundering (3 cycles)	4.0
Crocking	
Wet	3.0
Dry	4.0
Dimensional stability (3 cycles) (maximum)	
Wales	5%
Courses	5%
Fabric stretch (minimum)	
Wales	30%
Courses	50%
Vertical flame (initial)	
Wales and courses (maximum)	
After flame	2.0 seconds
Char length	5.0 inches
Melt/drip	No melt/no drip
Vertical flame (5 launderings)	
Wales and courses (maximum)	
After flame	2.0 seconds
Char length	5.0 inches
Melt/drip	No melt/no drip
Visual shade matching	
Daylight D75 (7500±200K)	Good match to standard sample
Incandescent (2856±200K)	Good match to standard sample
Water sorption/wicking	
Initial	Pass
After 20 launderings	Pass

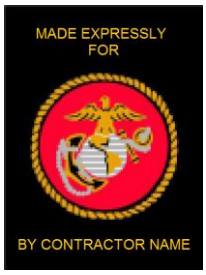
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3.3.2 Components.

3.3.2.1 Thread. The thread for the pullover shirt shall be aramid, Tex size 40 bobbin/looper conforming to A-A-55217. As an alternate, a para-aramid spun staple thread, Tex size 39 conforming to A-A-55195, type I, may be used. The color of the thread shall be Coyote 498.

3.3.2.2 Labels.

3.3.2.2.1 United States Marine Corps (USMC) exclusive label. Each pullover shirt shall have a woven USMC exclusive label, cut single, fused edge, 33 millimeters x 50 millimeters (1.299 inches x 1.969 inches), Bell Label, or equal. The exclusive label shall be placed inside at the center back of the neck. The top edge shall be caught in collar setting seam and the label stitched on all four sides. The information on the label shall include the following: "Made Expressly for" at the top, followed by a woven USMC emblem in the center, the contractor's name below the emblem, and the abbreviated size and length designation at the bottom of the label in the configuration below. All lettering shall be in gold.



3.3.2.2.2 Flame resistant organizational gear (FROG) label. Each pullover shirt shall have a cut single, fused edge, woven FROG label, 63 millimeters x 63 millimeters (2.480 inches x 2.480 inches) in size, manufactured by IbisTek, Bell Label, or equal. The FROG label shall be sewn on the outside of each pullover shirt at the left lower front (as worn) below the pocket opening and centered between rows of stitching. The bottom of the label shall be directly above the hem. The label shall be in the following configuration:



3.3.2.2.3 Combination size, identification, and instruction label. Each pullover shirt shall have a combination identification and instruction label. The label shall be in accordance with type VI, class 14 of MIL-DTL-32075 when tested as specified in 4.4.1. The color of the label shall be white. The label shall be caught in the stitching of the left side seam approximately 2½ inches from the hem. The combination label shall be facing the body and shall not be visible from the outside when the pullover shirt is in use. The combination size, identification, and instruction label shall include the following information and be in the following configuration:

Size and identification information:

PULLOVER SHIRT, FR MIDWEIGHT

CONTRACT NO:

NSN:

FIBER CONTENT:

CONTRACTOR'S NAME:

SIZE:

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Instruction Information:

Machine wash in lukewarm water (105 °F)

Tumble dry low or line dry

Do Not Use Fabric Softeners

DO NOT BLEACH, DRY CLEAN, OR IRON

3.3.2.2.4 Barcode label/tag. Each pullover shirt shall be individually barcoded with a label/tag conforming to type VI, class 17 of MIL-DTL-32075 when tested as specified in 4.4.1. The barcode label/tag shall be located so that it is completely visible on the item when it is folded or packaged as specified (see 6.2) and shall cause no damage to the item.

3.3.2.3 Slide fastener. The slide fastener for the front opening of the pullover shirt shall be a continuous polyester mono-filament element, size 5-6, conforming to type I, style 7 of A-A-55634 when tested as specified in 4.4.1. The color of the slide fastener shall be Coyote 498. The slider shall have an opening large enough to accommodate a ⅜-inch wide thong. Slide fastener lengths for the front opening are 14 inches for all sizes.

3.3.2.3.1 Webbing. The webbing used to reinforce the end of the slide fastener on the pullover shirt shall be 1-inch wide, 4 panel, spun polyester webbing conforming to Elastic Cord and Webbing, Inc. Style # 4056, or equal. The color of the webbing shall be Coyote 498. The webbing shall be no melt/no drip. Cut edges of webbing shall be turned under.

3.4 Design. The pullover shirt shall have long sleeves with a thumbhole in the cuff for donning and a collar with a reinforcing outer collar that converts into a mock turtleneck when the front opening slide fastener is completely closed. The slide fastener front opening shall extend from the top of the collar to the mid-chest and shall be backed with a full-length throat protection flap. The base of the front opening shall be reinforced with a 1- x 1½-inch piece of webbing. A large hand warmer pocket backed with a lighter weight pocketing liner shall extend from the lower chest area to the base of the pullover. The back of the pullover shirt shall be shaped to be longer than the front. All joining seams shall be flat locked and all hems shall be cover stitched (see figure 1 and figure 2).

3.4.1 Figures. Figures 1 and 2 are furnished for informational purposes only. To the extent an inconsistency exists between the written specification and the figures, the written specification shall govern.

3.5 Patterns. Standard patterns, providing a seam allowance of ⅜ inch for all seams except ½ inch for setting of zipper, will be furnished by the Government (see 6.2). The Government patterns shall not be altered in any way, and shall only be used only for cutting the contractor's working patterns. The working patterns shall be identical to the Government patterns, except that additional notching, if needed to facilitate manufacture is permitted on the contractor's working pattern. Minor modifications on the working patterns are permitted where necessary to accommodate the manufacturer's processes and the use of automated equipment provided the modifications do not alter the dimensional, serviceability, or appearance requirements cited in this document or those governed by the Government patterns.

3.5.1 Pattern parts. Component parts shall be cut from the materials specified and in accordance with the pattern parts listed in table IV.

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TABLE IV. List of pattern parts.

Material	Piece Name	Cut Parts
Basic material (3.3.1.1)	Back	1
	Upper front	1
	Bottom front	1
	Side lower front	2
	Top collar	1
	Side lower back facing	2
	Sleeve	2
	Sleeve facing	2
	Zipper protective flap	1
Under collar, zipper protective flap backing and lower front facings (3.3.1.2)	Under collar	1
	Side lower front facing	2
	Zipper protective flap lining	1
Pocket lining (3.3.1.3)	Bottom front pocket	1

3.6 Construction. The following specifics will provide garments with uniform appearance, comfort and durability for use during field duties, combat, and operations other than war. End item construction and appearance shall conform to the requirements and figures provided and the finished measurements cited (see 3.7) to maintain end item configuration.

3.6.1 Seams and stitching. The seams and stitching used shall be consistent, exhibit a uniform appearance, and conform to ASTM D6193 and the requirements for the location, stitch, and seam types listed in table V. Seam allowances shall be maintained with seams sewn so that no raw edges, runoffs, twists, pleats, puckers, or open seams result. All seams shall start and finish evenly and edges shall not unravel. Edges may be turned-in, turned-under, or serged to prevent raveling. Thread tension shall be maintained so that there is no tight, loose, or unbalanced stitching. The stitches per inch shall be 10-14 for all seams except stitch type 607 which shall be 12-14 stitches per inch.

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TABLE V. Seams and stitching.

Seaming Areas and Operations	Stitch Type	Seam Type
Join shoulders	607	FSa-1
Join edges of zipper flap, turn and topstitch with ¼-3⁄16 inch topstitch	301	SSe-2
Overedge sides and bottom edge of side lower back facing and raw edges of zipper flap	504	EFd-1
Attach side lower back facings to bottom front pocket around all sides	301	LSbj-1
Join lower front facings and lower front panel, turn and topstitch 3⁄16"- ¼" from folded edge of pocket opening. Turn under back edge of facing and stitch to lower front ⅛- 3⁄16 inch from turned edge	301	LSct-2
Join bottom front to bottom front pocket	301	SSa-1
Join side lower fronts to completed bottom front	607	FSa-1
Join top collar and undercollar, and overedge bottom of top collar	504	SSa-1
Turn down collar joining seam and raise stitch	301	SSa-1
Set collar catching USMC/size label at center back	607	FSa-1
Sew USMC/size label down around remaining edges	301	SSa-1
Set zipper with flap into pullover front opening, close collar, and topstitch collar and zipper 3⁄16-¼ inch from folded edge	301	LSbu-2
Attach reinforcement webbing at base of zipper opening	301	Box Stitch
Join completed upper front to completed bottom front	607	FSa-1
Bind thumbhole openings	402	BSc-2
Set sleeves flat	607	FSa-1
Close side seams, catching care label in left side seam 2-3 inches above bottom edge of pullover	607	FSa-1
Turn sleeve hems to outside ½ inch and hem	605	EFa-2
Turn bottom hem under ¾ inch and hem	605	EFa-2
Attach FROG label to lower front left side between seam joining lower front to side lower front, and pocket facing seam, bottom edge of label aligned directly above top edge of bottom hem seam	301	LSbj-1

3.6.2 Bartacks. Bartacks for reinforcement shall be placed in the length, quantity, and location as specified in table VI and as shown in figure 1 and figure 2.

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TABLE VI. Bartack locations.

Bartack Locations	Length (Inches)	Quantity Per Garment
Sleeve hem at inseam	⅝	2
End of bottom hem	⅝	1
Hand warmer pocket openings – 1 at top and bottom of each opening	⅝	4
Thumb hole opening – 1 at top and bottom of each opening	⅝	4

3.6.3 Tacking and backstitching. Ends of seams and rows of stitching with 301 type stitch, when not caught in other seams or stitching, shall be securely backstitched not less than ¼ inch. Thread breaks (all stitch types) shall be secured by stitching back of the break not less than ½ inch.

3.7 Finished dimensions. The finished pullover shirt shall conform to the measurements listed in table VII as defined in table VIII. All measurements shall be taken with the garment laid flat. Sleeve lengths shall not be uneven by more than ½ inch.

TABLE VII. Finished measurements (inches).

Regular	X-Small	Small	Medium	Large	X-Large	Tolerance
½ chest	19	21	23	25	27	±½
Back length	29	29½	30	30½	31	±¾
Sleeve length	25¾	26⅛	26½	26⅞	27¼	±½
Collar width at center front (CF)	3¼	3¼	3¼	3¼	3¼	±¼
Long			**Medium**	**Large**	**X-Large**	
½ chest			23	25	27	±½
Back length			32	32½	33	±¾
Sleeve length			27½	27½	28	±½
Collar width at CF			3¼	3¼	3¼	±¼

TABLE VIII. Methods of measure.

Measurement	Method
½ chest	Measure from side seam to side seam at base of armhole.
Back length	Measure from base of collar joining seam down center back to bottom of pullover hem.
Sleeve length	Measure from shoulder seam along the outer folded edge of sleeve to bottom of hem/binding.
Collar width at CF	Measure from bottom of collar seam to top of collar along right side of zipper attachment seam. Measure with the garment zipped and placed flat on back.

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3.8 Toxicity. The pullover shirt shall not present a health hazard and shall show compatibility with prolonged direct skin contact when tested as specified in 4.4.5. Chemicals recognized by the Environmental Protection Agency (EPA) as human carcinogens shall not be used. When specified (see 6.2), the contractor shall furnish information, which certifies that the finished product is composed of materials, which have been safely used commercially or provided sufficient toxicity data to show compatibility with prolonged, direct skin contact.

3.8.1 Toxicity documents. As specified (see 6.2), all finishes/chemicals used to process the garment shall be identified and accompanied by the appropriate Safety Data Sheet (SDS) information.

3.9 Workmanship. The finished pullover shirt shall be free from loose thread, foreign matter, and irregular defects that can adversely affect usage or durability. The finished pullover shirt shall conform to the quality of product established by this document, be uniform, and be free from defects specified in 4.4.3 and those defects that adversely affect form, fit, or function. Defects shall not exceed the Acceptable Quality Limits (AQLs) (see 6.2) and shall not adversely affect the serviceability, appearance, and uniformity of the product.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 First article inspection. A first article, in accordance with 3.1.1 shall be examined in accordance with 4.4.1 through 4.4.5 for compliance with design, configuration, workmanship, dimensional requirements, and testing as specified in this document (see 6.2).

4.2.1 First article samples and acceptance criteria. Unless otherwise specified in the contract (see 6.2), first article samples shall be provided. The sample size shall be as specified in the contract (see 6.2). The sample unit shall be one pullover shirt and the lot size shall be expressed in units of pullover shirts. The presence of excessive defects exceeding AQLs (see 6.2) or failure to pass any test shall be cause for rejection of the first article.

4.3 Conformance inspection. Unless otherwise specified in the contract (see 6.2), conformance inspection, in accordance with 3.1.2, shall consist of the examinations and tests as specified in 4.4.1 through 4.4.5.

4.3.1 Material and component sampling and acceptance criteria. Unless otherwise specified in subsidiary specifications, referenced documents, or procurement documents (see 6.2), sampling shall be in accordance with table IX. The lot shall be unacceptable if one or more sample units fail to meet any test requirement specified. All requirements are applicable to the sample unit. For material testing, the sample unit shall be 5 continuous yards full width of finished cloth for all physical and chemical tests.

TABLE IX. Material sampling.

Lot Size (Yards)	Sample Size (Sample Unit)
800 or less	2
801 through 22,000	3
22,001 and over	5

4.3.2 End item sampling and acceptance criteria. Unless otherwise specified in the contract (see 6.2) sampling for inspection shall be performed conforming to ANSI/ASQ Z1.4. The sample unit shall be one pullover shirt and the lot shall be expressed in units of pullover shirts. The presence of excessive defects (see 6.2) or failure of any testing as specified (see 6.2) shall be cause for rejection of the lot.

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4.4 Inspection methods.

4.4.1 Component and material examinations and tests. In accordance with 4.1, components and materials shall be inspected and tested in accordance with all the requirements of this specification and those referenced specifications and standards unless otherwise excluded, amended, or modified in this specification or applicable procurement documents (see 6.2). In addition to testing provisions specified in referenced documents, materials and components shall be examined and tested in accordance with table X through table XIV. All requirements are applicable to the sample unit. Unless otherwise specified (see 6.2), materials shall be conditioned and tested in accordance with ASTM D1776/D1776M. All requirements are applicable to the sample unit.

TABLE X. Component and material verification.

Material	Requirement	Verification
Basic material	3.3.1.1	As listed in table XI
Under collar, zipper protective flap backing and lower front facings	3.3.1.2	As listed in table XII
Pocket lining	3.3.1.3	As listed in table XIII
Thread	3.3.2.1	A-A-55217
Combination identification and instruction label	3.3.2.2.3	MIL-DTL-32075
Barcode label tag	3.3.2.2.4	MIL-DTL-32075
Slide fastener	3.3.2.3	A-A-55634
Webbing	3.3.2.3.1	As listed in table XIV

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TABLE XI. Basic material verification.

Material Characteristic	Material Requirement	Test Method
Construction	3.3.1.1	Visual
Fiber content	3.3.1.1	AATCC TM20A
Weight	3.3.1.1	ASTM D3776/D3776M
Air permeability	3.3.1.1	ASTM D737
Bursting strength	3.3.1.1	ASTM D3787
Colorfastness		
Laundering (3 cycles)	3.3.1.1	AATCC TM61, option 2A, grade polyester only
Crocking		
Wet	3.3.1.1	AATCC TM8
Dry	3.3.1.1	AATCC TM8
Light, Xenon	3.3.1.1	AATCC TM16.3, option 3 (85 kJ)
Dimensional stability (3 cycles)		
Wales	3.3.1.1	AATCC TM135, 3 cycles, (1) III, (A) ii
Courses	3.3.1.1	AATCC TM135, 3 cycles, (1) III, (A) ii
Fabric Stretch		
Wales	3.3.1.1	ASTM D2594 (Loose Fit)
Courses	3.3.1.1	ASTM D2594 (Loose Fit)
Pilling (jersey face)		
Initial	3.3.1.1	ASTM D3512/D3512M
After laundering (3 cycles)	3.3.1.1	AATCC TM135, (1) III, (A) ii; ASTM D3512/D3512M
Thermal insulation (Clo)		
Initial	3.3.1.1	ISO 11092
After 3 launderings	3.3.1.1	AATCC TM135, 3 cycles, (1) III, (A) ii, ISO 11092
Thickness	3.3.1.1	ASTM D1777
Vertical Flame (initial)		
Wales and courses		
After flame	3.3.1.1	
After glow	3.3.1.1	4.4.1.2, ASTM D6413/D6413M
Char length	3.3.1.1	
Melt/drip	3.3.1.1	
Vertical Flame (after 5 launderings)		AATCC TM135, 5 cycles, (1) III, (A) ii
Wales and courses		
After flame	3.3.1.1	
After glow	3.3.1.1	4.4.1.2, ASTM D6413/D6413M
Char length	3.3.1.1	
Melt/drip	3.3.1.1	
Visual shade matching		
Daylight (D75) (7500±200K)	3.3.1.1	AATCC EP9, option A
Incandescent (2856±200K)	3.3.1.1	AATCC EP9, option A

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TABLE XII. Under collar, zipper protective flap backing and lower front facings verification.

Material Characteristic	Material Requirement	Test Method
Construction	3.3.1.2	Visual
Fiber content	3.3.1.2	AATCC TM20A
Fiber denier	3.3.1.2	ASTM D1577
Fiber staple length	3.3.1.2	ASTM D5103
Yarn type		
Warp	3.3.1.2	Visual
Fill	3.3.1.2	
Weight	3.3.1.2	ASTM D3776/D3776M
Breaking strength		
Warp	3.3.1.2	ASTM D5034
Fill	3.3.1.2	
Char temperature (minimum)	3.3.1.2	Thermogravimetric analysis
Colorfastness		
Laundering	3.3.1.2	AATCC TM61, option 2A
Crocking		
Wet	3.3.1.2	AATCC TM8
Dry	3.3.1.2	AATCC TM8
Light, Xenon	3.3.1.2	AATCC TM16.3, option 3 (85 kJ)
Dimensional stability (3 cycles)	3.3.1.2	
Warp	3.3.1.2	AATCC TM135, (1) III, (A) ii
Filling	3.3.1.2	
Vertical flame (initial)	3.3.1.2	
Warp and fill		
After flame	3.3.1.2	ASTM D6413/D6413M
Char length	3.3.1.2	
Melt/drip	3.3.1.2	
Vertical flame (after 5 launderings)	3.3.1.2	AATCC TM135, 5 cycles, (1) III, (A) ii
Warp and fill		
After flame	3.3.1.2	
Char length	3.3.1.2	ASTM D6413/D6413M
Melt/drip	3.3.1.2	
Visual shade matching		
Daylight (D75) (7500±200K)	3.3.1.2	AATCC EP9, option A
Incandescent (2856±200K)	3.3.1.2	

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TABLE XIII. Pocket lining verification.

Material Characteristic	Material Requirement	Test Method
Construction	3.3.1.3	Visual
Fiber content	3.3.1.3	AATCC TM20A
Weight	3.3.1.3	ASTM D3776/D3776M
Bursting strength	3.3.1.3	ASTM D3787
Colorfastness (minimum)		
Laundering (3 cycles)	3.3.1.3	AATCC TM61, option 2A, grade polyester only
Crocking	3.3.1.3	
Wet	3.3.1.3	AATCC TM8
Dry	3.3.1.3	
Dimensional stability (3 cycles)		
Wales	3.3.1.3	AATCC TM135, 3 cycles, (1) III, (A) ii
Courses	3.3.1.3	
Fabric stretch		
Wales	3.3.1.3	ASTM D2594
Courses	3.3.1.3	
Vertical flame (initial)		
Wales and courses		
After flame	3.3.1.3	ASTM D6413/D6413M
Char length	3.3.1.3	
Melt/drip	3.3.1.3	
Vertical flame (5 launderings)		AATCC TM135, 5 cycles, (1) III, (A) ii
Wales and courses		
After flame	3.3.1.3	ASTM D6413/D6413M
Char length	3.3.1.3	
Melt/drip	3.3.1.3	
Visual shade matching		
Daylight (D75) (7500±200K)	3.3.1.3	AATCC EP9, option A
Incandescent (2856±200K)	3.3.1.3	AATCC EP9, option A
Water sorption/wicking		
Initial	3.3.1.3	4.4.1.1
After 20 launderings	3.3.1.3	AATCC TM135, 5 cycles, (1) III, (A) ii; 4.4.1.1

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TABLE XIV. Webbing verification.

Characteristic	Requirement	Test Method
Fiber identification	3.3.2.3.1	AATCC TM20A
Width	3.3.2.3.1	Visual/ruler
Melt/drip	3.3.2.3.1	ASTM D6413/D6413M
Visual shade matching		
Daylight (D75) (7500±200K)	3.3.2.3.1	AATCC EP9, option A
Incandescent (2856±200K)	3.3.2.3.1	AATCC EP9, option A

4.4.1.1 Water sorption and wicking. Water sorption and wicking shall be determined using the following procedure.

4.4.1.1.1 Water sorption and wicking test specimens. Fabric specimens shall be conditioned in accordance with ASTM D1776/D1776M and tested in that environment. The specimen size shall be 6 inches by 6 inches; three (3) separate specimens shall be used for each of the face side and back side tests. A fabric shall be considered passing only when tests on both the face side and the back side meet the respective test pass/fail criteria on all individual specimens tested.

4.4.1.1.2 Face side wicking test. The test specimen shall be laid flat on a glass plate with back side up (i.e., inner or skin surface when used in a garment). One (1) drop of 0.10±0.01 milliliter of distilled water at 70±2 °F shall be placed in the center of the test specimen using a pipettor and a stopwatch/timer immediately started. The test specimen shall then be immediately turned over on the glass plate with test specimen face side up. The diameter of the wicked water area (denoted by a darkened water mark) shall be measured at a total elapsed time of 10 seconds. The specimen shall be considered passing if the diameter of the wicked water area (darkened water mark) is equal to or greater than 1 3/16 inches.

4.4.1.1.3 Back side sorption test. The test specimen shall be laid flat on a glass plate with back side up (i.e., inner or skin surface when used in a garment). One (1) drop of 0.10±0.01 milliliter of distilled water at 70±2 °F shall be placed in the center of the test specimen using a pipettor and a stopwatch/timer immediately started. The water (denoted as a darkened water mark) shall be observed and the time for the water mark to disappear (water sorption, denoted as a lightened water mark approximating the shade of the basic material) shall be recorded. The specimen shall be considered passing if the water sorption (disappearance of the darkened water mark) is 10 seconds or less.

4.4.1.2 Modified vertical flame test. The vertical flame test shall be conducted on samples of the basic material and aramid cloth, which meets the requirements in table II. Both materials shall be cut in the same direction for each test. The basic material cloth shall be placed in the sample holder, face side facing up. The bottom edge (toward the flame) shall be folded back ½ inch. The back side of the fabric shall now be facing the flame. The aramid cloth shall be placed along the edge of the basic material cloth and secured in the sample holder. The vertical flame test shall be performed in accordance with ASTM D6413/D6413M.

4.4.2 In-process examination. Visual and dimensional examinations shall be made at any point or during any phase of the manufacturing process to determine whether construction details, which cannot be examined in the finished product, are in accordance with the requirements specified in section 3. Materials and components which contain defects and damages, as defined in the visual examination of 4.4.3, shall be removed from production. Any in-process nonconformance remaining in the finished pull over shirt shall be classified as a defect in accordance with the visual examination of 4.4.3.

4.4.3 Visual examination. The pullover shirt shall be examined for defects in color, shade, design, material, construction, workmanship, and markings, and the defects shall be classified in accordance with the list of defects specified in table XV. The inspection conditions and the AQLs shall be as specified (see 6.2).

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TABLE XV. Visual examination.

Examination	Defect	Classification	
		Major	Minor
Workmanship	Component part omitted, distorted, full, tight, or twisted; any part of end item caught in any unrelated stitching; the edge of any component part required to be forced out having folds of more than ⅛ inch.	101	
Material	Hole, cut, tear, smash, burn, drill hole, run, needle chew, visible mend, thin place, or misweave affecting appearance or serviceability.	102	
Shade	a. Shade variation within a part or between parts. Thread color not as specified.	103	
	b. Dye streak; color not as specified	104	
Seams and stitching	a. Seams puckered, distorted, pleated, wavy, or twisted; edge or raised stitching sewn too close to edge, resulting in damage to cloth.		201
	b. Irregular or open seam; raw edge affecting appearance or serviceability	105	
	c. Seam, seam allowance, or stitch type not as specified		202
	d. Loose or tight stitch tension; broken or missing thread or stitch		203
	e. Box X-stitching missing; stitches insecure, broken, misplaced, not specified size		204
Bartacks	Bartacks missing, insecure, misplaced, not specified size, stitches loose or broken.		205
Cleanliness	Spot, stain, excessive thread ends not trimmed or removed, odor.		206
Labels	a. Label omitted, incorrect type, size, or configuration; illegible, inaccurate, or not attached as specified	106	
	b. Barcodes omitted or not readable by scanner; human-readable interpretation (HRI) omitted or illegible		207
	c. Barcode not visible on folded, packaged item		208
	d. Barcode attachment causes damage to the item	107	
Packaging	Any pullover shirt not packaged in accordance with the contract or purchase order.		209

4.4.4 Dimensional examination. The pullover shirt shall be examined for conformance to the dimensions specified in table VII as defined in table VIII. Any measurement deviating from the dimensions and tolerances specified shall be scored as a measurement defect. The inspection conditions and the AQLs shall be as specified (see 6.2).

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4.4.5 Toxicity assessment. An acute dermal irritation study and a skin sensitization study shall be conducted on laboratory animals. When the results of the studies indicate the undershirt is not a sensitizer or irritant, a Repeat Insult Patch Test shall be performed in accordance with the Modified Draize Procedure (Hayes' Principles and Methods of Toxicology and Contact Allergy: Predictive Testing in Humans). If the toxicity requirement (see 3.8) can be demonstrated with historical data, then toxicity testing may not be required. At a minimum, toxicity data should include results from a primary dermal irritation study in laboratory animals and a repeated insult human patch test (Modified Draize Procedure). The latter must be conducted under the supervision of a qualified dermatologist using at least 100 free-living individuals.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The pullover shirt is intended for use with the cold weather clothing system worn by personnel of the USMC.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.

b. Sizes and lengths required (see 1.1).

c. National stock number (NSNs).

d. When first article is required (see 3.1.1, 4.2, and 6.3).

e. When conformance inspection is required (see 3.1.2 and 4.3).

f. Applicable Government patterns and reference standards, including revisions (see 3.3.1.1, 3.3.1.2, 3.3.1.3, and 3.5).

g. Barcode/label tag location (see 3.3.2.2.4).

h. Requirement for the submittal of information which certifies that the finished product is composed of materials which have been safely used commercially; or sufficient toxicity data to show compatibility with prolonged, direct skin contact (see 3.8).

i. Requirement for an SDS and the identification of all finishes/chemicals used to process the garment (see 3.8.1).

j. AQLs (see 3.9 and 4.4.4).

k. First article sampling and acceptance criteria (see 4.2.1).

l. Component and material sampling and acceptance criteria (see 4.3.1).

m. End item sampling and acceptance criteria (see 4.3.2).

n. Component and material examination and tests (see 4.4.1).

o. Inspection conditions (see 4.4.3 and 4.4.4).

p. Packaging requirements (see 5.1).

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6.3 First article. When first article inspection (see 6.2) is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be first article sample, a first production item, or a standard production item from the contractor's current inventory, and the number of items to be tested as specified in 4.2. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for pending contract. Bidders should not alternate submit bids unless specifically requested to do so in the solicitation.

6.4 Suggested sources.

a. Basic material and pocket lining:

(1) Polartec, LLC, 46 Stafford St., Lawrence, MA 01841, (978) 685-6341

b. Webbing:

(1) Elastic Cord & Webbing, Inc., 155 Prairie Lake Rd., Unit G, East Dundee, IL 60118, (847) 844-4122

c. Woven labels:

(1) Bell Label, 777 Main St., Lewiston, ME 04240, (207) 784-2961

(2) Ibis Tek, LLC, 912 Pittsburgh Rd., Butler, PA 16002, (724) 586-6005

6.5 NSNs. For informational purposes only, an NSN that corresponds to this specification is listed in table XVI. Table XVI is not intended to be a comprehensive list of all NSNs associated with this specification.

TABLE XVI. NSN.

NSN	Size
8415015553895	Medium Regular

6.6 Supersession data. This document supersedes MC/PD 06 2011 SYSCOM, dated 20 October 2011, in its entirety.

6.7 Subject term (key word) listing.

Grid fleece

Level 2

Marine Corps

Underwear

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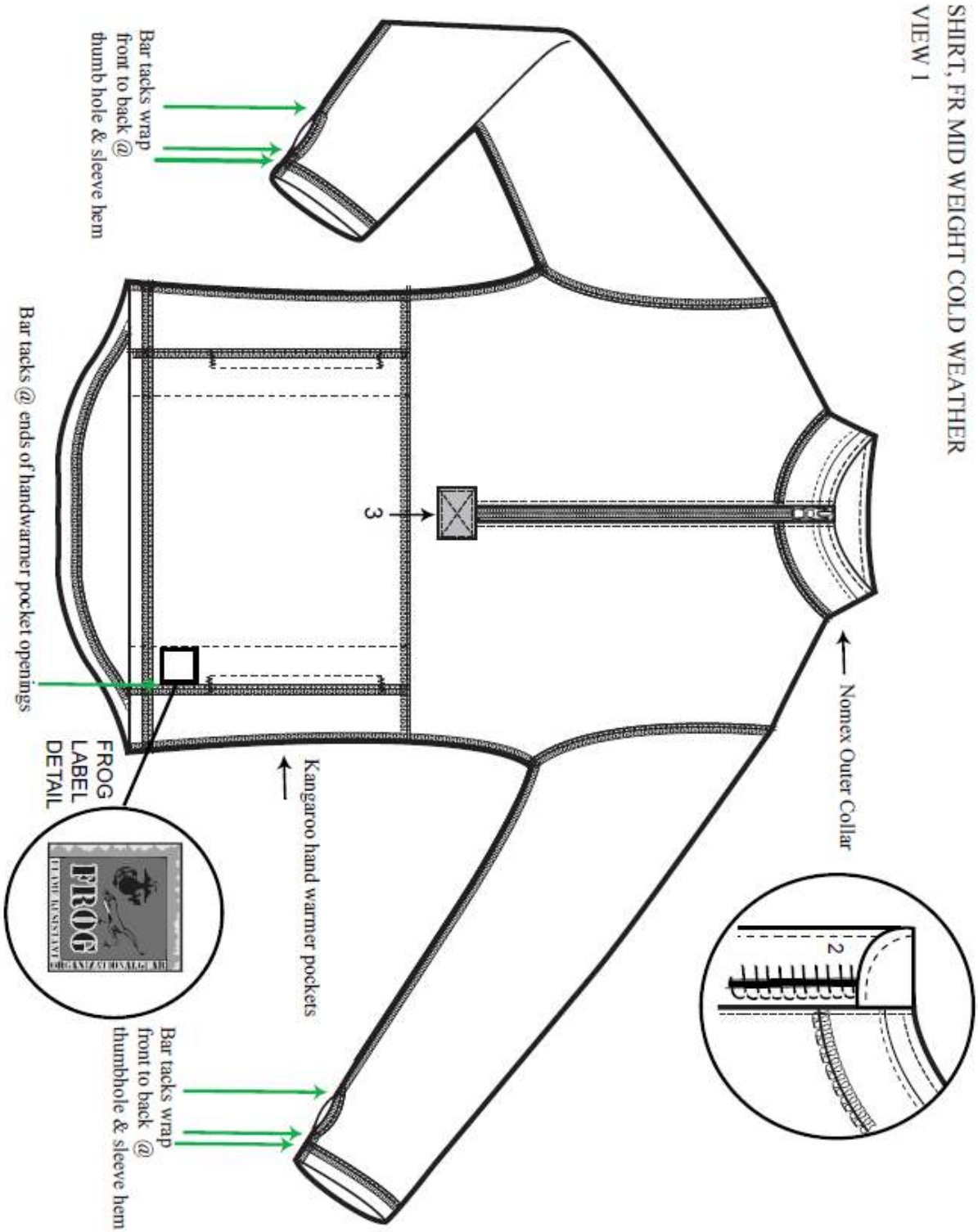


FIGURE 1. Front view.

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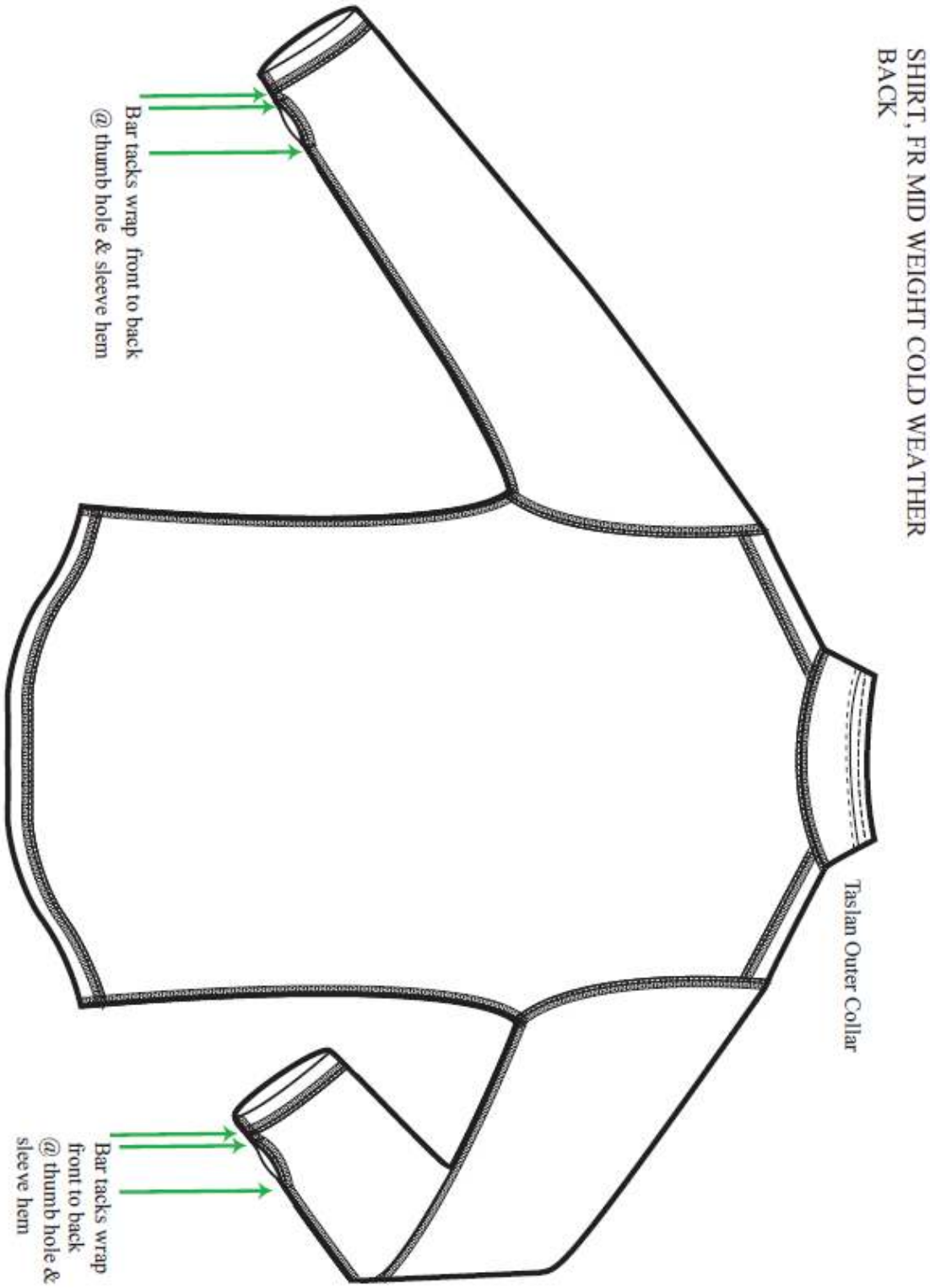


FIGURE 2. Back view.

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CONCLUDING MATERIAL

Custodians:
Army – GL
Navy – MC
Air Force – 11

Review activities:
Army – AV, MI
Navy – CG1, NU
Air Force – 03, 70
DLA – CT

Civil agency:
GSA – FAS

Preparing activity:
Navy – MC
(Project 8415-2018-004)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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